The spherically symmetric Einstein–Maxwell–dilaton Penrose inequality at $\alpha = 1$

Igor Itkin
Independent researcher
ig.itkin@gmail.com

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Abstract

The charged Penrose inequality tests cosmic censorship in the presence of matter. For the Einstein–Maxwell–dilaton (EMD) system at dilaton coupling $\alpha = 1$, Khuri, Weinstein and Yamada [8] conjectured the sharp bound $2M \geq \sqrt{\rho_H^2 + 2q^2}$, with M the ADM mass, ρ_H the horizon area radius and q the charge, saturated by the Gibbons–Maeda / Garfinkle–Horowitz–Strominger (GHS) black hole; the available positive-mass and spinorial methods reach only an additive charge term, not this Pythagorean form. We prove the conjecture for static, spherically symmetric, time-symmetric data satisfying the dominant energy condition, with equality if and only if the data is GHS. The proof uses a quasilocal mass that is nondecreasing along the area-radius foliation, equals $\frac{1}{2}\sqrt{\rho_H^2 + 2q^2}$ at the horizon and tends to M at infinity; its monotonicity reduces to a polynomial positivity certificate verified in exact arithmetic. We show in addition that the divergence-free hypothesis $\operatorname{div}(e^{-2\phi}E) = 0$ on the Maxwell field is necessary, whereas the sub-extremality condition $\rho_H \geq q$ often imposed on charged Penrose inequalities is redundant here, and we give a second, independent proof of the bound by a radical-free rational mass.

Keywords: Penrose inequality; Einstein–Maxwell–dilaton; dilaton black hole; quasilocal mass; spherical symmetry.

1 Introduction

The Penrose inequality asserts that the total mass M of an asymptotically flat spacetime is bounded below by the area A of its outermost horizon, $2M \geq \sqrt{A/4\pi}$; it is a sharpened, quantitative form of the cosmic censorship conjecture [1]. In the time-symmetric (Riemannian) case it was proved by Huisken and Ilmanen [5] for a single horizon and by Bray [6] for several, in each case with equality only for the Schwarzschild slice. When the matter carries electric charge q, a sharp charge-dependent refinement is expected, saturated by the Reissner–Nordström solution, and it has been established under a range of hypotheses [7, 9, 10]. These charged results test cosmic censorship precisely where it is most delicate: near extremality, where the horizon area is small relative to the charge.

The Einstein–Maxwell–dilaton (EMD) system couples the Maxwell field to a scalar dilaton ϕ through the action $S = \int(R - 2|\nabla\phi|^2 - e^{-2\alpha\phi}F^2)$, where F is the Maxwell two-form and α the dilaton coupling; it arises in the low-energy limit of string theory and in Kaluza–Klein reduction. At $\alpha = 1$ its canonical black hole is the GHS solution [4, 3], and (Section 2) it saturates the *Pythagorean* bound

$$2M \;\geq\; \sqrt{\rho_H^2 + 2q^2}.$$

Khuri, Weinstein and Yamada [8] conjectured that this bound holds for all admissible EMD data, under a divergence-free hypothesis on the Maxwell field. The problem we solve is exactly their conjecture in spherical symmetry.

Conjecture 1.1 (Khuri–Weinstein–Yamada [8])**.** *Let the static, spherically symmetric EMD data of Section 2 at coupling $\alpha = 1$ satisfy the dominant energy condition, with a connected outermost minimal-surface horizon of area radius ρ_H, conserved charge q, and $\operatorname{div}(e^{-2\phi}E) = 0$ outside the horizon. Then $2M \geq \sqrt{\rho_H^2 + 2q^2}$, with equality if and only if the data is GHS.*

The difficulty is that the sharp right-hand side is Pythagorean in (ρ_H, q), whereas the classical routes to mass lower bounds in this setting (the Witten–Nester spinor identities, which give the positive mass theorem with charge $M \geq |q|/\sqrt{2}$ [14, 15, 16], and static uniqueness at $\alpha = 1$ [17]) produce an *additive* boundary term of the form $q/\sqrt{2} + \rho_H/2$. That additive term cannot reproduce $\sqrt{\rho_H^2 + 2q^2}$ except near equality (it exceeds it by a triangle-inequality defect), so these methods bound or rigidify the mass without yielding a Penrose inequality, and the sharp bound has remained open.

In this paper we prove Conjecture 1.1 for spherically symmetric data, and we clarify which of its hypotheses are essential. The main result, proved in Section 5, is the following.

Theorem 1.2 (Spherical EMD Penrose inequality, $\alpha = 1$; [8])**.** *Let (M^3, g, E, ϕ) be static, spherically symmetric, asymptotically flat EMD data at $\alpha = 1$ satisfying the dominant energy condition, with a connected outermost horizon of area radius ρ_H, conserved charge q, and source-free Maxwell field $\operatorname{div}(e^{-2\phi}E) = 0$. Then*

$$2M \;\geq\; \sqrt{\rho_H^2 + 2q^2},$$

with equality if and only if the data is the GHS black hole (3).

The source-free hypothesis is not a technical convenience but a genuine requirement, and Section 3 shows precisely what its absence costs.

Theorem 1.3 (Necessity of the source-free hypothesis)**.** *For every $\rho_H > 0$ and every $q \neq 0$ there is admissible static, spherically symmetric, DEC EMD datum, carrying an interior charge source, with outermost horizon of area radius ρ_H and conserved charge q, whose ADM mass is arbitrarily close to $\rho_H/2$. Hence once $\operatorname{div}(e^{-2\phi}E) = 0$ is dropped, no bound $B(\rho_H, q) > \rho_H/2$ holds; in particular $\frac{1}{2}\sqrt{\rho_H^2 + 2q^2}$ fails.*

By contrast, a *further* side condition often attached to charged Penrose inequalities (the sub-extremality bound $\rho_H \geq q$, imposed in the Einstein–Maxwell case to exclude over-charged data [7, 9, 10]) turns out to be unnecessary here.

Proposition 1.4 (Redundancy of $\rho_H \geq q$)**.** *The proof of Theorem 1.2 uses no upper bound on $\sigma_H := \sqrt{1 + 2q^2/\rho_H^2}$. Consequently $2M \geq \sqrt{\rho_H^2 + 2q^2}$ holds for every admissible static, spherically symmetric, DEC datum, including those with $\rho_H < q$.*

Finally, we prove the bound twice, by two monotone masses of different algebraic type: the charged Hawking–Bray mass used for Theorem 1.2, whose one nested radical is removed by uniformization, and a *rational*, radical-free mass built by Hermite interpolation on the GHS locus (Section 7). Because one mass carries a radical and the other is rational, the second proof is an independent check on the first.

Remark 1.5 (The uncharged case, for every α)**.** At $q = 0$ there is no Maxwell field, the dilaton is harmonic, and the DEC reduces to $R_g = 2|\nabla\phi|^2 \geq 0$. The bound is then an immediate corollary of the ordinary Riemannian Penrose inequality [5, 6]: $2M \geq \rho_H$, with equality iff Schwarzschild, for every coupling α. The dilaton coupling enters only through the charged sector, which is the content of Theorem 1.2.

Remark 1.6 (The right-hand side is coupling-dependent)**.** The form $\sqrt{\rho_H^2 + 2q^2}$ is special to $\alpha = 1$: already on the GHS family it fails for $\alpha > 1$ (for instance $\alpha = 2$). The correct sharp per-coupling bound is the GHS-calibrated $B_\alpha(\rho_H, q)$ read off the GHS mass–radius relation, of which $\frac{1}{2}\sqrt{\rho_H^2 + 2q^2}$ is the $\alpha = 1$ case. The present paper treats $\alpha = 1$.

For static EMD black holes the positive mass theorem with charge $M \geq |q|/\sqrt{2}$ (spinor methods [14, 15, 16]) and static uniqueness at $\alpha = 1$ [17] are known; as noted above, neither carries a horizon-area term, so neither is a Penrose inequality. The closest *proven* Penrose inequalities in adjacent settings are the spherically symmetric charged case without dilaton (Einstein–Maxwell, and with a Gauss–Bonnet term) of Kunduri, Margalef-Bentabol and Muth [11], and the Riemannian Penrose inequality for weighted manifolds [12] (uncharged, with a weighted-static rather than GHS equality case); neither covers the charged EMD bound. Nozawa, Shiromizu, Izumi and Yamada [13] obtain a Penrose-*type* inequality in the EMD system through the divergence/positive-mass route, but not the sharp $2M \geq \sqrt{\rho_H^2 + 2q^2}$ with GHS rigidity established here.

The paper is organized as follows. Section 2 fixes the data and the GHS solution, and Section 3 proves Theorem 1.3. Section 4 outlines the strategy for Theorem 1.2, which is carried out in Section 5. Section 6 proves Proposition 1.4, and Section 7 gives a second, independent proof via a rational mass.

2 Setup and the GHS solution

This section fixes the initial-data class, writes the dominant energy condition as the single scalar constraint used in the proof, and records the GHS solution that saturates the bound.

2.1 Initial data

We work with a static, spherically symmetric, time-symmetric initial data set for the EMD system at coupling $\alpha = 1$. In area-radius gauge the spatial metric is

$$g_3 = \frac{\mathrm{d}\rho^2}{1 - \frac{2m(\rho)}{\rho}} + \rho^2\,\mathrm{d}\Omega^2, \tag{1}$$

so ρ is the areal radius, $\mathrm{d}\Omega^2$ the unit round metric on S^2, and $m(\rho)$ the Misner–Sharp mass [2], characterized by $1 - 2m/\rho = |\nabla\rho|^2$. The ADM mass is $M = \lim_{\rho\to\infty} m(\rho)$; the conserved Maxwell charge is $q = \frac{1}{4\pi}\oint e^{-2\phi} \star F$, the same on every sphere by Gauss' law; and ρ_H is the horizon area radius, i.e. the outermost minimal surface, where $m(\rho_H) = \rho_H/2$. Throughout, ϕ denotes the dilaton and E the electric field. The dominant energy condition (DEC) is assumed.

2.2 The dominant energy constraint

For such data the DEC is a single scalar constraint on the profiles $m(\rho), \phi(\rho)$. Writing $u := 2m/\rho \in [0,1)$ for the compactness and $p := \rho\,\phi'(\rho)$ for the dilaton momentum, the Misner–Sharp mass obeys

$$2m'(\rho) = \frac{q^2 e^{2\phi}}{\rho^2} + (1-u)\,p^2 + 2\varepsilon, \qquad \varepsilon \geq 0, \tag{2}$$

where the first term is the Maxwell energy density, the second the dilaton gradient energy, and $\varepsilon \geq 0$ collects any further DEC-respecting matter. The GHS black hole has $\varepsilon \equiv 0$.

2.3 The GHS solution

The canonical solution is the GHS black hole [4, 3], written in a Schwarzschild-like radius r with parameters $r_p > r_m > 0$:

$$e^{2\phi} = 1 - \frac{r_m}{r}, \qquad \rho(r)^2 = r(r - r_m), \qquad r_+ = r_p, \qquad q^2 = \tfrac{1}{2} r_p r_m, \qquad M = \tfrac{1}{2} r_p. \tag{3}$$

Its horizon area radius is $\rho_H = \rho(r_p) = \sqrt{r_p(r_p - r_m)}$, so that $\rho_H^2 + 2q^2 = r_p(r_p - r_m) + r_p r_m = r_p^2$, and therefore

$$2M = r_p = \sqrt{\rho_H^2 + 2q^2}. \tag{4}$$

Thus GHS saturates the bound of Theorem 1.2, which is what makes the Pythagorean right-hand side the natural sharp form and GHS the expected equality case.

3 Necessity of the source-free hypothesis

3.1 The charge-shell counterexample

Without the source-free hypothesis the bound of Theorem 1.2 fails by an arbitrary factor: the mass can be driven down to the uncharged value $\rho_H/2$ while the charge q is held fixed and made arbitrarily large. We prove Theorem 1.3 by an explicit construction.

Proof of Theorem 1.3. Carry the full charge q on a thin dilatonic shell at radius ρ_s, with an uncharged interior minimal surface of area radius $\rho_H < \rho_s$. By Gauss' law the horizon then encloses no charge: it is a Schwarzschild horizon, and the Riemannian Penrose inequality $m \geq \rho_H/2$ [5, 6] is saturated to leading order. The only mass above $\rho_H/2$ is the exterior field energy $\int_{\rho_s}^{\infty} \frac{1}{2} e^{-2\phi}|E|^2\rho^2\,\mathrm{d}\rho = O(q^2/\rho_s)$, which tends to 0 as $\rho_s \to \infty$. Concretely, $\rho_H = 1$, $\phi \equiv 0$, $q = 5$ ramped onto a shell at $\rho_s \approx 500$ gives $M_{\mathrm{ADM}} = 0.5247$, against $\frac{1}{2}\sqrt{1 + 2\cdot 25} = \frac{1}{2}\sqrt{51} = 3.57$, a violation by a factor 6.8; letting $\rho_s \to \infty$ at fixed (ρ_H, q) drives $m \to \rho_H/2$. $\square$

3.2 The screening mechanism

This is the EMD transcription of McCormick's correction [10] to the charged Riemannian Penrose inequality (the $\alpha = 0$ case): a charge screened from the horizon by a distant shell contributes to the asymptotic q but not to the mass, because at the shell $\operatorname{div}(e^{-2\phi}E) \neq 0$, so the horizon never encloses it. The divergence-free hypothesis of Conjecture 1.1 excludes exactly this configuration, which is why it is essential rather than technical.

4 Outline of the proof

The proof of Theorem 1.2 is carried out in Section 5 and is a monotonicity argument for a single quasilocal mass. We construct in Section 5.1 a mass $G(\rho) = \rho\, H_{\mathrm{br}}$, the charged Hawking–Bray mass, calibrated to the GHS reference flow so that $G(\rho_H) = \frac{1}{2}\sqrt{\rho_H^2 + 2q^2}$ at the horizon and $G(\rho) \to M$ at infinity (Section 5.5). The inequality then follows from a single fact: that G is nondecreasing along the area-radius foliation.

Monotonicity is reduced to pointwise algebra on each leaf. Differentiating G and using the dominant energy constraint (2) writes $\mathrm{d}G/\mathrm{d}\rho$ as a quadratic in the dilaton momentum $p = \rho\phi'$, the flow identity of Section 5.2; hence $\mathrm{d}G/\mathrm{d}\rho \geq 0$ for every profile is equivalent to two inequalities in the leaf variables—positivity of the leading coefficient, $H_u > 0$, and nonpositivity of the discriminant, $\mathrm{Disc} \leq 0$ (Lemma 5.1).

The positivity $H_u > 0$ is direct. The discriminant inequality is the crux, and it is obstructed by a single nested radical $W = \sqrt{R}$ carried by H_{br}: in the original variables Disc admits no sum-of-squares representation. In Section 5.4 we remove the radical by uniformizing the conic $W^2 = R$, after which $\mathrm{Disc} \leq 0$ becomes a genuine polynomial positivity, certified in exact arithmetic by a Bernstein and Sturm argument. Equality forces the double-root locus $R = 1$; tracing it back reproduces the GHS solution and gives the rigidity statement (Section 5.6).

Two further developments accompany the main proof. Section 6 shows that the certificate is uniform in the charge-to-radius ratio, so the sub-extremality hypothesis $\rho_H \geq q$ is never used (Proposition 1.4), and Section 7 gives a second, independent proof of the bound with a radical-free rational mass, which checks the first.

5 The monotone quasilocal mass

We now carry out the proof of Theorem 1.2 along the plan of Section 4, constructing the mass and establishing in turn its flow identity, the two leaf inequalities, the endpoint values, and rigidity.

5.1 Definition of the mass

The proof of Theorem 1.2 exhibits a quasilocal mass $G(\rho)$ that is nondecreasing along the radial foliation, equals the right-hand side of the inequality at the horizon, and tends to M at infinity. We work in three dimensionless leaf variables,

$$t = e^{\phi}, \qquad u = \frac{2m}{\rho}, \qquad \sigma = \sqrt{1 + \frac{2q^2}{\rho^2}}, \tag{5}$$

where t is the dilaton scale, $u \in [0, 1)$ the compactness of (2), and $\sigma \geq 1$ the charge-to-radius ratio, together with the single nested radical

$$z = \sigma t, \qquad \kappa = \frac{4t^2(1-u)(1-t^2)}{(t^2+1)^2}, \qquad R = z^2 + \kappa, \qquad W = \sqrt{R}. \tag{6}$$

The monotone mass is $G = \rho\, H_{\mathrm{br}}(t, u, \sigma)$ with

$$H_{\mathrm{br}}(t, u, \sigma) = \frac{(t^2-1)^2 + 4u\,t^2}{2t\,(t^2+1)^2} + \frac{W-1}{2t} - \frac{2t\,(1-u)\,(R-1)^2}{(W+z)\,(z+1)^2\,(t^2+1)^2}. \tag{7}$$

This is the charged Hawking–Bray mass. It is rational in (t, u, σ) apart from the radical W; it is constant along the GHS reference flow (Section 5.6) and nondecreasing on every other dominant-energy profile, as we now show.

5.2 The flow identity

Write $H = H_{\mathrm{br}}$ and let subscripts denote its partial derivatives. Differentiating $G = \rho\, H$ along the foliation and using (2) gives the exact flow identity

$$\frac{\mathrm{d}G}{\mathrm{d}\rho} = A_2\, p^2 + A_1\, p + A_0 + 2H_u\, \varepsilon, \tag{8}$$

with coefficients

$$\begin{aligned} A_2 &= (1-u)\, H_u, \qquad A_1 = t\, H_t, \\ A_0 &= H - u\, H_u - \frac{\sigma^2-1}{\sigma}\, H_\sigma + \frac{\sigma^2-1}{2}\, t^2\, H_u. \end{aligned} \tag{9}$$

Identity (8) is the chain rule. Differentiating the leaf variables (5) along the foliation gives

$$\rho\, t' = t\, p, \qquad \rho\, \sigma' = -\frac{\sigma^2-1}{\sigma}, \qquad \rho\, u' = 2m' - u,$$

and substituting the constraint (2) and $q^2/\rho^2 = (\sigma^2-1)/2$ into $\rho\, u'$, then collecting powers of p, yields exactly the coefficients (9). The identity holds for arbitrary radial profiles $m(\rho), \phi(\rho)$.

Since (8) is a quadratic in the momentum p with a nonnegative ε-term, monotonicity $\mathrm{d}G/\mathrm{d}\rho \geq 0$ reduces to two statements:

$$H_u > 0 \quad (\Rightarrow A_2 = (1-u)H_u > 0 \text{ and } 2H_u\varepsilon \geq 0), \qquad \mathrm{Disc} := A_1^2 - 4A_2A_0 \leq 0. \tag{10}$$

Lemma 5.1 (Monotonicity of G). *On the admissible range $t \in (0,1),\ u \in [0,1),\ \sigma \geq 1$ one has $H_u > 0$ and $\mathrm{Disc} \leq 0$, with $\mathrm{Disc} = 0$ if and only if $W = 1$. Hence by (8) every DEC profile satisfies $\mathrm{d}G/\mathrm{d}\rho \geq 0$.*

We prove Lemma 5.1 in the next two subsections.

5.3 Positivity of H_u

Differentiating (7) in u and using $W^2 = R$ to clear the radical, one finds that H_u is a positive multiple of a rational expression whose denominator is $t\,D_2 + \sigma(t^2+1)^2 W$, with

$$D_2 = \sigma^2(t^2+1)^2 + 2(t^2-1)(u-1).$$

Here $D_2 > 0$ on the whole range: for $t < 1$ the term $2(t^2-1)(u-1)$ is a product of two nonpositive factors, hence ≥ 0, so $D_2 \geq \sigma^2(t^2+1)^2 \geq (t^2+1)^2 > 0$. The numerator is positive on the entire range as well: an exact polynomial positivity, certified by the same Bernstein and Sturm argument used for the discriminant cofactor in Section 5.4. No point of $\{t \in (0,1),\ u \in [0,1),\ \sigma \geq 1\}$ has $H_u \leq 0$, so $H_u > 0$ unconditionally, for every $\sigma \geq 1$ with no upper cap; in particular there is no threshold at $\sigma = \sqrt{3}$ (see Section 6).

5.4 Nonpositivity of the discriminant

Exact computation factors the discriminant as

$$\mathrm{Disc} = A_1^2 - 4A_2A_0 = -(W-1)^2\, P(t,u,\sigma), \tag{11}$$

with P the cofactor left after extracting $-(W-1)^2$. Since $(W-1)^2 \geq 0$, the inequality $\mathrm{Disc} \leq 0$ (with equality if and only if $W = 1$) is equivalent to $P > 0$ on the admissible range, which we establish below. That $R \geq 0$, so that W is real, is itself transparent from the identity

$$R\,(t^2+1)^2 = t^2\Big[(t^2-1+2u)^2 + 4(1-u)(1+u) + (\sigma^2-1)(t^2+1)^2\Big], \tag{12}$$

whose bracket is a sum of three summands, each nonnegative on $0 \leq u \leq 1,\ \sigma \geq 1$. It remains to certify $P > 0$. A direct sum-of-squares argument is impossible: in the monomial basis P has thousands of sign-indefinite coefficients, because P still carries the radical $W = \sqrt{R}$. We remove it by uniformizing the conic $W^2 = R$, after which positivity becomes a basis-level statement.

First we uniformize. With κ and z as in (6), set

$$\xi := z + W, \qquad \text{so} \qquad W = \frac{\xi^2+\kappa}{2\xi}, \qquad \sigma = \frac{\xi^2-\kappa}{2t\,\xi}. \tag{13}$$

Then W and σ are rational in (t,u,ξ), and $W^2 = R$ holds identically. The map $z \mapsto \xi = z + W$ is monotone, since $\mathrm{d}\xi/\mathrm{d}z = 1 + z/W > 0$, so the physical region $\{z > 0,\ W > 0\}$ is exactly the half-line $\xi \geq \xi_0$, where $\xi_0 := t + s$ and $s := \sqrt{t^2+\kappa} > 0$; the radicand $t^2[(t^2-1)^2 + 4 + 4u(t^2-1)]$ of s is linear in u and positive at both $u = 0$ and $u = 1$.

Next we factor out the rigidity locus. In the ξ-chart the discriminant is the rational identity

$$\mathrm{Disc} = \frac{t^2\,(1-u)\,L^2\,C(t,u,\xi)}{8\,\xi^6\,(t^2+1)^{12}\,F_1F_2^2F_3^6}, \qquad L = 2\xi\,(W-1)\,(t^2+1)^2, \tag{14}$$

where

$$F_1 = -2\xi z\,(t^2+1)^2, \qquad F_2 = 2\xi W\,(t^2+1)^2, \qquad F_3 = -2\xi(z+1)\,(t^2+1)^2. \tag{15}$$

The numerator minus Disc times the denominator vanishes identically; the factor $L^2 \propto (W-1)^2$ is precisely what produced the $(W-1)^2$ of (11). On the physical region $z, W > 0$ force $F_1 < 0$, $F_2 > 0$, $F_3 < 0$, so the denominator $8\xi^6(t^2+1)^{12}F_1F_2^2F_3^6 < 0$, and $L^2 \geq 0$ isolates the rigidity factor ($L = 0 \Leftrightarrow W = 1$). Hence

$$\operatorname{sign}\mathrm{Disc} = -\operatorname{sign}C(t,u,\xi) \qquad \text{on } \{\xi \geq \xi_0\}, \tag{16}$$

and the sign question is reduced to the positivity of the polynomial cofactor C.

Finally we certify $C \geq 0$ on the half-line. Expanding the cofactor about the regional endpoint ξ_0,

$$C(t,u,\xi) = \sum_{k=0}^{24} \frac{P_k(t,u) + Q_k(t,u)\,s}{(t^2+1)^{24-k}}\,(\xi-\xi_0)^k. \tag{17}$$

Since $\xi \geq \xi_0$, every $(\xi-\xi_0)^k \geq 0$; the top coefficient is $(t^2+1)^{22} > 0$ (with $Q_{24} = 0$), so C is strictly positive at leading order. Each P_k, Q_k is a polynomial in the two variables (t,u) alone—the radical and σ are gone. We certify $P_k, Q_k \geq 0$ on $\{t > 0,\ 0 \leq u \leq 1\}$ by Bernstein positivity [18, 19]: a polynomial written in the Bernstein basis $B_{n,j}(u) = \binom{n}{j}u^j(1-u)^{n-j}$ over $[0,1]$ is nonnegative there as soon as all its Bernstein coefficients $b_j(t)$ are nonnegative on $t > 0$, since the basis is itself nonnegative. Rewriting each P_k, Q_k in this basis produces the one-variable coefficient family $\{b_j(t)\}$; each $b_j(t)$ is certified nonnegative on $(0,\infty)$ in exact integer arithmetic: immediately whenever its t-coefficients are all nonnegative, and otherwise by a Sturm root-count (no real root in $(0,\infty)$, together with a positive value at $t = 1$). (The companion positivity of the H_u numerator in Section 5.3 is certified the same way, after splitting the t-axis into the charts $\{0 < t \leq 1\}$ and $\{t \geq 1\}$ (the latter via $t \mapsto 1/t$) so that each b_j lies on a bounded interval.) Hence $P_k, Q_k \geq 0$, so $C \geq 0$ by (17), and $\mathrm{Disc} \leq 0$ by (16), with equality if and only if $W = 1$, i.e. on GHS. This proves Lemma 5.1. □

The certificate is exact throughout: the Bernstein reduction and each Sturm root-count [20] are decided over $\mathbb{Z}$, so it is checkable coefficient by coefficient. Because the radical makes the positivity of Disc inaccessible to any hand or elimination argument in the original variables, the uniformization is essential: it removes the radical and reduces the sign question to the polynomial cofactor C, to which the Bernstein/Sturm certificate applies.

The certificate is also independent of the charge regime. In (6)–(13) the charge variable enters C only through $z = \sigma t$, hence through ξ; the expansion (17)–(16) lives in (t,u,ξ) on the half-line $\xi \geq \xi_0$ with no upper bound, so $\mathrm{Disc} \leq 0$ holds for all $\sigma \geq 1$ at once. Likewise the horizon coefficient

$$A_0\big|_{u=1} = \frac{(\sigma^2t^2-1)^2}{2\sigma(t^2+1)^2} \;\geq 0 \qquad (\text{all } \sigma) \tag{18}$$

is a manifest square, with no cap on σ. These facts are used in Section 6.

5.5 The endpoints and proof of the main theorem

At the horizon, $u \to 1$. The third term of (7) carries the factor $(1-u)$ and drops; at $u=1$ one has $\kappa = 0$, $R = z^2$, $W = z = \sigma t$, and $(t^2-1)^2 + 4t^2 = (t^2+1)^2$, so

$$H_{\mathrm{br}}(t, 1, \sigma) = \frac{(t^2+1)^2}{2t(t^2+1)^2} + \frac{\sigma t - 1}{2t} = \frac{1}{2t} + \frac{\sigma t - 1}{2t} = \frac{\sigma}{2}. \tag{19}$$

Hence, with $\sigma_H = \sqrt{1 + 2q^2/\rho_H^2}$,

$$G(\rho_H) = \rho_H\, H_{\mathrm{br}}(t, 1, \sigma_H) = \frac{\rho_H\, \sigma_H}{2} = \frac{1}{2}\sqrt{\rho_H^2 + 2q^2}, \tag{20}$$

which is the right-hand side of Theorem 1.2, valid identically in σ.

At infinity the data flatten: $t \to 1$, $u = 2m/\rho \to 0$, $\sigma \to 1$. One computes $H_{\mathrm{br}}(1,0,1) = 0$ and $H_u(1,0,1) = \frac{1}{2}$, so $H_{\mathrm{br}} \simeq H_u\, u = M/\rho$ and

$$G(\infty) = \lim_{\rho\to\infty} \rho\, H_{\mathrm{br}} = M. \tag{21}$$

Proof of Theorem 1.2. Combining the flow identity (8), Lemma 5.1, and the endpoints (20) and (21),

$$M = G(\infty) = G(\rho_H) + \int_{\rho_H}^{\infty} \frac{\mathrm{d}G}{\mathrm{d}\rho}\,\mathrm{d}\rho \;\geq\; G(\rho_H) = \frac{1}{2}\sqrt{\rho_H^2 + 2q^2}.$$

The equality statement is proved in Section 5.6. □

5.6 Rigidity

Equality in Theorem 1.2 forces $\mathrm{d}G/\mathrm{d}\rho \equiv 0$, hence $\varepsilon \equiv 0$ and the quadratic $A_2p^2 + A_1p + A_0$ vanishes identically. As $A_2 > 0$ and $\mathrm{Disc} \leq 0$, this requires $\mathrm{Disc} = 0$ at the double root $p = -A_1/(2A_2)$; by (11), $\mathrm{Disc} = 0 \Leftrightarrow W = 1 \Leftrightarrow R = 1$. Tracing the locus $R = 1$ back through (5)–(6) reproduces the GHS relations (3), along which one verifies directly that

$$R \equiv 1, \qquad G = \rho\, H_{\mathrm{br}} \equiv \frac{r_p}{2} = M \quad \text{identically in } \rho.$$

Thus G is constant exactly on GHS, and equality holds only there. This completes the proof of Theorem 1.2.

6 Redundancy of the sub-extremality hypothesis

6.1 The condition and its redundancy

The source-free and connected-outermost hypotheses of Theorem 1.2 are necessary; Section 3 shows what their absence costs. A *further* side condition is often attached to charged Penrose inequalities, by analogy with the Einstein–Maxwell case, where over-charged data can violate the bound and a sub-extremality assumption is imposed [7, 9, 10]; in the present setting it reads

$$\sigma_H \leq \sqrt{3} \iff \rho_H \geq q \iff r_p \geq \tfrac{3}{2} r_m \tag{22}$$

(the last equivalence holding on GHS). Unlike the source-free and connected hypotheses, this one is never used in the proof of Theorem 1.2: that is the content of Proposition 1.4, which we now prove.

Proof of Proposition 1.4. Both ingredients of the proof are regime-free. First, the horizon endpoint (19) gives $H_{\mathrm{br}}(t, 1, \sigma) = \sigma/2$ for all σ, so (20) yields $G(\rho_H) = \frac{1}{2}\sqrt{\rho_H^2 + 2q^2}$ with no restriction on σ_H, consistent with the manifest square (18). Second, by the closing paragraphs of Section 5.4 the certificate of Lemma 5.1 proves $H_u > 0$ and $\mathrm{Disc} \le 0$ for all $\sigma \ge 1$; nothing degrades as σ crosses $\sqrt{3}$. (Directly, Disc stays strictly negative arbitrarily far into $\sigma > \sqrt{3}$: for instance $\mathrm{Disc} = -18.0$ at $\sigma = 5$, $t = 0.8$, $u = 0.6$, with $A_2 = 0.50 > 0$ and $P = 1.98$ in (11).) Both endpoints and the monotonicity therefore hold without (22), and the chain $M = G(\infty) \ge G(\rho_H) = \frac{1}{2}\sqrt{\rho_H^2 + 2q^2}$ is valid for $\rho_H < q$ as well. □

6.2 Admissibility and explicit competitors

The one genuine structural requirement is the standard one for any Penrose inequality: ρ_H must be the outermost minimal surface, i.e. $u < 1$ for all $\rho > \rho_H$. This is regime-independent and is exactly the mechanism that protects the bound. If one over-concentrates mass near the horizon (large positive ϕ_H, or a large near-horizon ε), the compactness re-enters $u > 1$ at some $\rho > \rho_H$: a trapped surface forms outside, ρ_H is no longer outermost, and the datum is inadmissible. Only there, where the trajectory leaves the box $u \in [0, 1)$, can $\mathrm{d}G/\mathrm{d}\rho$ turn negative. For admissible data with $\rho_H < q$ the constraint is simply that the dilaton be sufficiently negative at the horizon, $e^{2\phi_H} < \rho_H^2/q^2$, which GHS satisfies automatically. Integrating (2) outward from $u(\rho_H) = 1$ with non-GHS dilaton profiles and $\varepsilon \ge 0$ produces explicit DEC data with $\rho_H < q$, and all admissible ones obey the bound: for instance $\rho_H = 1$, $q = 2$, $\sigma_H = 3$ gives $2M = 3.05$ against $\sqrt{\rho_H^2 + 2q^2} = 3.00$, a margin of 1.7%.

Remark 6.1 (Why $\rho_H \ge q$ is nonetheless quoted)**.** The condition (22) is load-bearing for the spinorial (Witten-type) proof, not for the present quasilocal-flow proof. As noted in Section 1, the spinor route produces an additive boundary term $\sim q/\sqrt{2} + \rho_H/2$, which cannot reproduce the Pythagorean $\sqrt{\rho_H^2 + 2q^2}$ except near equality; the gap is a triangle-inequality defect controlled precisely by $\sigma_H \le \sqrt{3}$. The condition was a feature of that method, mistaken for a feature of the theorem. Separately, the value $\sigma = \sqrt{3}$ once appeared as an apparent threshold in a closed-form expression for the partials of H_{br}; that threshold was an artifact of differentiating at fixed W rather than along $W = \sqrt{R}$, and with the chain term restored it disappears, so $H_u > 0$ holds unconditionally (Section 5.3). The flow proof never forms the additive quantity and never needs the regime.

7 A second proof via a radical-free mass

The mass H_{br} of Section 5 carries the nested radical $W = \sqrt{R}$, whose positivity forced the uniformization of Section 5.4. We now give a second, independent proof of Theorem 1.2, using a different monotone mass, rational in (t, u, σ), with no radical. Its monotonicity certificate is a direct Bernstein positivity, needing no uniformization. Because one mass carries a radical and the other is rational, their agreement is a strong check against computational error.

7.1 Construction

We build the mass by interpolating its profile in the leaf variable u between two distinguished leaves (the horizon $u = 1$ and the GHS locus) with data fixed so that G reproduces the GHS mass. The interpolant is a rational function of (t, u, σ), so no radical enters.

The interpolation data comes from two quantities. The u-linear part of the radical mass H_{br} is

$$\gamma = \frac{(t^2-1)^2 + 4ut^2}{2t(t^2+1)^2}, \tag{23}$$

and the GHS locus is the zero set of $\mathcal{P}$, at which u takes the value u_G:

$$\mathcal{P}(t,u,\sigma) = 4t^2(1-t^2)(1-u) - (t^2+1)^2(1-\sigma^2t^2), \qquad u_G = 1 - \frac{(t^2+1)^2(1-\sigma^2t^2)}{4t^2(1-t^2)}. \tag{24}$$

On the locus the mass must equal $\widehat{m} = \gamma|_{u=u_G}$ with prescribed u-slope b; together with an auxiliary rational function k these are

$$k = \frac{2t^2(\sigma+t)}{(t^2+1)^2(\sigma t+1)}, \qquad b = \frac{t}{t^2+1}. \tag{25}$$

Hermite interpolation in u (matching the horizon value $\frac{\sigma}{2}$ at $u = 1$ and the locus data $(\widehat{m}, b)$ at $u = u_G$) gives a base interpolant $\widehat{G}_4$. A single rational correction completes it to the mass profile $\widehat{G}_9$, whose monotonicity is certified in Section 7.3:

$$\widehat{G}_4 = \tfrac{\sigma}{2} + (u-1)(2k-b) - (u-1)^2\,\frac{b-k}{1-u_G}, \qquad \widehat{G}_9 = \widehat{G}_4 + \frac{2(b-k)^2}{b+k}\,\frac{(u-1)(u-u_G)^2}{(1-u_G)^2}. \tag{26}$$

The quasilocal mass is $G = \rho\,\max\!\big(\tfrac{u}{2},\, \widehat{G}_9\big)$. On the complement of the active set $\mathcal{A} = \{\widehat{G}_9 \geq u/2\}$ the branch $G = \rho u/2 = m$ is monotone directly by (2), and the maximum of two monotone functions is monotone.

7.2 Design identities

Write Δ_9 for the discriminant of the flow quadratic (8) generated by $\widehat{G}_9$ (its explicit form is given in (28) below). The interpolant satisfies, identically in (t, σ), the design identities

$$\widehat{G}_9(t,1,\sigma) = \tfrac{\sigma}{2}, \quad \widehat{G}_9|_{u=u_G} = \widehat{m}, \quad \partial_u\widehat{G}_9|_{u=u_G} = b, \quad \Delta_9|_{u=1} = 0, \quad \Delta_9|_{u=u_G} = 0, \tag{27}$$

together with $G \equiv m$ along the GHS slice (all verified directly). The horizon identity again gives $G(\rho_H) = \frac{1}{2}\sqrt{\rho_H^2 + 2q^2}$, and the two vanishings of Δ_9, at the horizon and on the locus, encode the double-root tangency that rigidity requires.

7.3 The rational certificate

The flow identity (8) holds verbatim for any $F = F(t,u,\sigma)$, so on $\mathcal{A}$ one has $\mathrm{d}G/\mathrm{d}\rho = A_2^s p^2 + A_1^s p + A_0^s + 2\beta_9\,\varepsilon$ with $\beta_9 = \partial_u \widehat{G}_9$, $A_2^s = (1-u)\beta_9$, and $\Delta_9 = (A_1^s)^2 - 4A_2^s A_0^s$. Writing num for the numerator over the common denominator (which is manifestly positive on $\mathcal{D} = \{t > 0, 0 < u < 1, \sigma > 1\}$),

$$\operatorname{num}(\beta_9) = t\,B_9, \qquad \operatorname{num}(\Delta_9) = (u-1)\,t\,\mathcal{P}^2\,M_9, \tag{28}$$

with $\mathcal{P}$ the GHS-locus polynomial above, and B_9 (57 monomials, multidegree $(16,2,4)$ in (t,u,σ)) and M_9 (472 monomials, multidegree $(35,3,11)$) polynomials. Since $u-1<0$ and $\mathcal{P}^2 \geq 0$ on $\mathcal{D}$, the two monotonicity conditions reduce to the polynomial inequalities

$$B_9 \geq 0 \quad \text{and} \quad M_9 \geq 0 \qquad \text{on } \mathcal{D}. \tag{29}$$

Both inequalities are low-degree in u (B_9 is quadratic, M_9 cubic), so expanding each in the Bernstein basis $B_{n,k}(u) = \binom{n}{k}u^k(1-u)^{n-k}$ over $[0,1]$ leaves only its Bernstein coefficients, polynomials in (t,σ); since $B_{n,k} \geq 0$ on $[0,1]$, nonnegativity of every coefficient gives the inequality on $\mathcal{D}$.

For M_9 the coefficients are certified directly: with $\sigma = 1+w$, $w>0$,

$$(1+t)^8\,M_9 = \sum_{k=0}^{3}\Big(\sum_{i,j\geq 0} c_{ij}^{(k)}\,t^i w^j\Big)\,B_{3,k}(u), \qquad c_{ij}^{(k)} \geq 0, \tag{30}$$

an identity in which every coefficient is nonnegative [18, 19]; as $t, w > 0$ and $B_{3,k} \geq 0$, it exhibits $M_9 \geq 0$ on $\mathcal{D}$ by inspection.

For $B_9 = b_0(1-u)^2 + 2b_1u(1-u) + b_2u^2$ each coefficient carries the physical variable σt through the factor $(\sigma t+1)$ and completes a square in σ. The top one is a one-line certificate: $b_2 = (t^2+1)^4(\sigma t+1)^2\,r_2$ with

$$4C_2\,r_2 = (2C_2\,\sigma + C_1)^2 + 28\,t^2(t^2-1)^2(t^2+1)^2, \qquad C_2 = t^2\big((t^2-2)^2+7\big) > 0, \tag{31}$$

so $r_2 > 0$ for every σ (here $C_1 = -4t(t^2-2t-1)(t^2+2t-1)$). The remaining coefficients are b_0 and $b_1 = (t^2+1)^2(\sigma t+1)\,r_1$, where $r_1 = b_1/[(t^2+1)^2(\sigma t+1)]$ is a polynomial in (t,σ). Expanding each in $w = \sigma - 1$ and writing the lowest, quadratic, parts as $c_0 + c_1 w + c_2 w^2$ (for r_1) and $d_0 + d_1 w + d_2 w^2$ (for b_0), the terms of order w^3 and higher have positive coefficients, while the quadratic parts have $c_0, c_2 > 0$, $d_0, d_2 > 0$ and manifestly nonpositive discriminant,

$$c_1^2 - 4c_0c_2 = -t^2(t+1)^4\,p_{16}, \qquad d_1^2 - 4d_0d_2 = -4t^2(t+1)^6(t^2+1)^2\,p_{20}, \tag{32}$$

with cofactors p_{16}, p_{20} nonnegative on $t > 0$ (Appendix A). Hence $b_0, b_1, b_2 \geq 0$, so $B_9 \geq 0$ on $\mathcal{D}$.[1] Hence $\mathrm{d}G/\mathrm{d}\rho \geq 0$, and the proof of Theorem 1.2 runs as in Section 5.5 with H_{br} replaced by $\widehat{G}_9$. Unlike H_{br}, this certificate is manifestly polynomial: no radical, and hence no uniformization step.

[1] On the extremal slice $\sigma = 1$ the scaffold collapses to one-variable squares in $\tau = t + t^{-1} \geq 2$; for instance $r_2|_{\sigma=1} = t^2(t+1)^2\big((\tau-3)^2+7\big)$.

8 Concluding remarks

We have proved the spherically symmetric Einstein–Maxwell–dilaton Penrose inequality at $\alpha = 1$ (Theorem 1.2), the sharp Pythagorean bound $2M \geq \sqrt{\rho_H^2 + 2q^2}$ conjectured by Khuri, Weinstein and Yamada, with GHS rigidity. We have shown that the divergence-free hypothesis on the Maxwell field is necessary (Theorem 1.3), whereas the sub-extremality condition $\rho_H \geq q$ is redundant (Proposition 1.4). The bound is proved twice, by two monotone masses of different algebraic type (the radical Hawking–Bray mass and the rational mass of Section 7), so that each proof checks the other.

For non-spherical data the analogous monotonicity is open and is pursued elsewhere; the current evidence is a convergent numerical flow with no violation down to $\rho_H/q = 0.63$. The present paper settles the spherically symmetric case and removes the charge–horizon hypothesis from it.

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Data availability

The manuscript source and a machine-verifiable supplementary bundle (a `sympy` script that re-derives, in exact rational arithmetic, every algebraic claim of Section 7.3 and Appendix A, together with the polynomials B_9, M_9 and the Handelman certificate for M_9) are available at https://github.com/YehudaItkin/spherical-emd-penrose.

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A The cofactors p_{16} and p_{20}

Through (32), the nonnegativity of b_0 and b_1 in Section 7.3 rests on the positivity for $t > 0$ of the two univariate cofactors

$$\begin{aligned} p_{16}(t) &= 3t^{16} - 24t^{15} + 36t^{14} + 536t^{13} - 2148t^{12} + 3736t^{11} - 1828t^{10} - 3512t^{9} \\ &\quad + 9938t^{8} - 9256t^{7} + 5788t^{6} - 2200t^{5} + 2044t^{4} + 424t^{3} - 540t^{2} + 56t + 19, \qquad (33) \\ p_{20}(t) &= 2t^{20} - 18t^{19} + 17t^{18} + 404t^{17} - 1085t^{16} - 856t^{15} + 5444t^{14} + 4032t^{13} - 27688t^{12} \\ &\quad + 35508t^{11} - 3266t^{10} - 22936t^{9} + 15170t^{8} + 11240t^{7} - 12492t^{6} + 2336t^{5} \\ &\quad + 3102t^{4} - 562t^{3} - 199t^{2} + 36t + 3. \qquad (34) \end{aligned}$$

The identities (32) need only that p_{16} and p_{20} be nonnegative for $t > 0$, and in fact both are strictly positive there. By Sturm's theorem [20], neither p_{16} nor p_{20} has a root in $(0, \infty)$; since the values $p_{16}(1) = 3072$ and $p_{20}(1) = 8192$ are positive, each polynomial is positive on the whole of $(0, \infty)$. The root count is exact, in integer arithmetic.

The only object in the proof too large to print is the identity (30), which writes $(1 + t)^8 M_9$ (the 472 monomials of M_9 cleared by $(1 + t)^8$) as a sum of 1844 nonnegative terms; these are provided as an ancillary file and were generated and checked in Wolfram Language. Every identity and positivity statement in Section 7.3 and here is exact over $\mathbb{Q}$, and the two inequalities $B_9, M_9 \geq 0$ are independently reconfirmed on $\mathcal{D}$ by cylindrical algebraic decomposition [20].